

Chapter 3: Representing and Interpreting Data

A Level Statistics Notes · Rexponential

Why this chapter matters

After summarising data numerically in Chapter 2, this chapter turns to visual representations: histograms for grouped continuous data with unequal class widths, cumulative frequency curves and box plots for finding quartiles graphically, and how to compare two distributions properly. Exam papers frequently combine a numerical measure from Chapter 2 with a diagram from this chapter in the same question, so the two chapters are meant to be used together rather than in isolation.

3.1 Histograms

Concept

A histogram represents grouped continuous data. Unlike a bar chart, it is not the height of each bar that shows the frequency — it is the **area**. This matters as soon as class widths are unequal: if a wide class and a narrow class have the same frequency, the wide bar must be drawn shorter, otherwise its larger area would make its frequency look bigger than it really is. So instead of plotting frequency directly on the vertical axis, histograms plot the **frequency density**.

Key vocabulary

Term	Meaning
Class boundary	The true upper/lower limit of a class (halfway between adjacent classes)
Class width	The difference between the upper and lower class boundaries
Frequency density	Frequency ÷ class width — the height plotted on a histogram

Key formula

KEY FORMULA

$$\text{Frequency density} = \frac{\text{Frequency}}{\text{Class width}}$$

Procedure

1. Convert class limits to class boundaries if needed (e.g. "10–19" becomes 9.5–19.5).
2. Calculate the class width for each class.
3. Calculate the frequency density for each class using the formula above.
4. Plot bars with the class boundaries on the horizontal axis and frequency density as the height — bars must be drawn touching, with no gaps, since the variable is continuous.

Working backwards from a histogram. This is a very common question type: use one bar where both the frequency and the area are known to find the *scale* (the frequency represented by each unit of area, or by each cm^2 if the histogram is a printed diagram), then apply that same scale to find the frequency of any other bar from its area.

Worked example

EXAMPLE 3.1

The table shows the times, in minutes, taken by 90 runners to complete a race.

Time (min)	30–40	40–45	45–50	50–70
Frequency	20	25	30	15

(a) Calculate the frequency density for each class. (b) On the histogram, the bar for the 45–50 class has a height of 6. Find the height of the bar for the 50–70 class.

Step 1 (part a). Class widths (upper boundary – lower boundary): 30–40 $\rightarrow$ 10, 40–45 $\rightarrow$ 5, 45–50 $\rightarrow$ 5, 50–70 $\rightarrow$ 20. **M1**

Frequency density: $20/10 = 2$, $25/5 = 5$, $30/5 = 6$, $15/20 = 0.75$

Step 2 (part b). The height given for the 45–50 bar (6) is exactly its frequency density (also 6, from part (a)), so this histogram's height axis is frequency density itself — a 1:1 scale. This means the height of any other bar is simply its frequency density. **M1**

Height of 50–70 bar = 0.75

Answer (a) Frequency densities are 2, 5, 6 and 0.75. (b) The 50–70 bar has height 0.75.

Exam tip

EXAM TIP

Exam papers lean heavily on the *reverse* process — given a histogram (or a partially completed one) and one known frequency, find another frequency using the ratio of areas — rather than simply asking you to draw a histogram from a table. Make sure you practise reading a scale from one known bar and applying it to others, not just the forward calculation shown above.

3.2 Cumulative frequency diagrams and box plots

Concept

A **cumulative frequency diagram** plots the running total of frequency against the *upper* class boundary of each class. Plotting cumulative frequency lets us read off, or calculate, the median and quartiles graphically — for example, the point where the running total reaches 50% of the data marks the median.

A **box plot** summarises a data set using its **five-number summary** — minimum, lower quartile, median, upper quartile, maximum — drawn as a box (spanning the interquartile range, with a line at

the median) and two whiskers. When outliers are present, they are usually plotted as separate points, and the whisker is drawn only as far as the most extreme value that is not an outlier.

Key vocabulary

Term	Meaning
Cumulative frequency	The running total of frequencies up to and including a class
Upper class boundary	The point plotted on the horizontal axis for each cumulative frequency value
Five-number summary	Minimum, Q1, median, Q3, maximum
Whisker	The line from the box to the minimum/maximum (or to the last non-outlier value)

Key formula (quartile by linear interpolation)

KEY FORMULA (QUARTILE BY LINEAR INTERPOLATION)

$$Q_k = L + \left(\frac{\frac{kn}{4} - F}{f} \right) \times h$$

$k = 1, 2$ or 3 , for the lower quartile, median, or upper quartile respectively

L = the lower class boundary of the class containing that quartile

F = the cumulative frequency up to (but not including) that class

f = the frequency of that class

h = the width of that class

Procedure

1. Build a cumulative frequency table, plotting each cumulative total against the *upper* class boundary.
2. Either read the median and quartiles directly from a smooth curve through these points (graphical method), or apply the interpolation formula directly to the table (numerical method) — use whichever the question asks for.
3. Combine the quartiles with the minimum and maximum to give the five-number summary.
4. Draw the box plot to scale: a box from Q1 to Q3 with a line at the median, whiskers to the minimum and maximum (or to the last non-outlier value if outliers are being shown separately), and outliers plotted individually.

Worked example

EXAMPLE 3.2

40 students' times (seconds) to complete a puzzle are summarised below.

Time (s)	< 10	< 20	< 30	< 40	< 50
Cumulative frequency	0	6	18	32	40

Estimate the median and the interquartile range.

Step 1. Median position: $n/2 = 40/2 = 20$, which falls in the class 30–40 (cumulative frequency rises from 18 to 32), so $L = 30$, $F = 18$, $f = 14$, $h = 10$. **M1**

$$\text{Median} = 30 + \left(\frac{20 - 18}{14} \right) \times 10 = 31.4$$

Step 2. For Q_1 : position = $n/4 = 40/4 = 10$, which falls in class 20–30, so $L = 20$, $F = 6$, $f = 12$, $h = 10$. For Q_3 : position = $3n/4 = 3 \times 40/4 = 30$, which falls in class 30–40, so $L = 30$, $F = 18$, $f = 14$, $h = 10$. **M1**

$$Q_1 = 20 + \left(\frac{10 - 6}{12} \right) \times 10 = 23.3$$

$$Q_3 = 30 + \left(\frac{30 - 18}{14} \right) \times 10 = 38.6$$

Step 3. $IQR = 38.6 - 23.3 = 15.3$. **A1**

Answer Estimated median ≈ 31.4 seconds; estimated IQR ≈ 15.3 seconds.

Exam tip

EXAM TIP

When a question explicitly asks for a graphical method, mark schemes accept any answer read from a correctly drawn curve, within a small tolerance — so how accurately the curve is drawn matters just as much as how carefully the value is read off it. When no graph is required, use the interpolation formula directly from the table instead, which gives an exact, examinable value, as shown above.

3.3 Comparing distributions

Concept

Comparing two data sets (e.g. two classes' test scores, or two locations' temperatures from the large data set) always requires **both** a measure of location and a measure of spread — stating only one gives an incomplete picture, since two data sets can share a median but have very different spreads, or vice versa. Every comparison must also be written **in context**: naming the actual variable and what a higher or lower value means in the real situation, not just which number is bigger.

Key vocabulary

Term	Meaning
Comparative statement	A sentence comparing two data sets on both location and spread
Context	Referring to the real-world variable and situation, not just abstract numbers

Procedure

1. Compare a measure of location (median or mean) between the two data sets, stating which is higher/lower **in context**.
2. Compare a measure of spread (IQR or standard deviation) between the two data sets, again in context — a larger spread means more variability/inconsistency in that variable.
3. If relevant, comment on skewness or any outliers shown by box plots.
4. Link the comparison back to the original question: what does this mean for the situation being studied?

Worked example

EXAMPLE 3.3

Box plots summarise the times (minutes) taken to solve a puzzle by two groups. Group A: minimum 4, $Q1 = 8$, median = 12, $Q3 = 15$, maximum 20. Group B: minimum 6, $Q1 = 9$, median = 10, $Q3 = 11$, maximum 16. Compare the two groups' performance.

Step 1. Compare location: Group A's median (12) is higher than Group B's median (10), so typically Group B solved the puzzle faster than Group A. **B1**

Step 2. Compare spread: Group A's $IQR = 15 - 8 = 7$, Group B's $IQR = 11 - 9 = 2$. Group A's times are far more spread out (more variable) than Group B's, which were fairly consistent. **B1**

Step 3. Interpretation in context: Group B not only solved the puzzle faster on average but did so more consistently, while Group A's performance varied much more widely from student to student. **B1**

Answer Group B typically completed the puzzle faster (median 10 vs 12 minutes) and with much less variation between students ($IQR\ 2$ vs 7 minutes) than Group A.

Exam tip

EXAM TIP

The single most commonly dropped mark in comparison questions is failing to write the comparison **in context** — stating " $12 > 10$ " earns nothing on its own, but "Group A took longer, on average, to solve the puzzle than Group B" does. Drill this explicitly: every comparative sentence should name the variable and say what the difference actually means, not just report which number is bigger.

Chapter summary

Idea	Key point
Histogram	Height = frequency density = frequency $\div$ class width; area, not height, represents frequency
Cumulative frequency diagram	Plot cumulative frequency against the <i>upper</i> class boundary
Quartile interpolation	$Q_k = L + ((kn/4 - F)/f) \times h$
Box plot	Five-number summary: minimum, Q1, median, Q3, maximum, with outliers plotted separately if present
Comparing distributions	Always compare both location and spread, in context

Common exam mistakes

- Plotting a histogram's height as frequency instead of frequency density when class widths are unequal.
- Plotting cumulative frequency against the *lower* or *midpoint* class boundary instead of the upper boundary.
- Comparing only location or only spread between two data sets, rather than both.
- Writing a numerical comparison without referring to the real-world context ("12 > 10" instead of what that difference means).

You should now be able to

- Draw and interpret a histogram with unequal class widths, including finding an unknown frequency from a given scale.
- Construct a cumulative frequency diagram and use it (or the interpolation formula) to estimate the median and quartiles.
- Construct a box plot from a five-number summary, including outliers.
- Write a full comparison between two distributions, covering both location and spread, in context.